SUPPLEMENTARY DOCUMENT

Table E-1. Summary of recent systematic reviews and meta-analyses

Article	Databases	Date of search	Inclusion Criteria	Major Outcomes Analysed	Number of Studies	Major Results (Odds [CI95%, Heterogeneity])*	Subgroups Analyzed	Study Quality
Lodge 2015, Acta Paediatrica	PubMed, CINAHL and EMBASE	July 2015	Observational (cohort and non-cohort) and experimental studies published in full length in English with wheeze at any age and/or asthma >= 5 years of age. Excluded studies on premature births. Included outcomes were spirometric or physician diagnosed asthma including parental report or from database and reported asthma or wheeze.	Asthma or wheeze	29	Ever vs. Never 0.88 (0.82, 0.95) I^2 = 44% Exclusive >3-4 vs. <3-4 months 0.89 (0.71, 1.11) I^2 = 72% More vs. Less 0.90 (0.84, 0.97) I^2 = 63%	age (5-10, >10), study size, year at birth, study design, length of breastfeeding, control for confounding, country affluence	NOS/Grade Score very low to low
Dogaru 2014, American Journal of Epidemiology	PubMed and EMBASE	December 2013	Fully reported original observational studies (cohort and non-cohort) published in English, excluding studies in special populations (e.g. atopic history) and those exploring only asthma severity. Included outcomes are physician-diagnosed asthma including parental report, treatment for asthma/wheezing, and parental reports of wheezing in the last 12mo.	Asthma ever, current asthma	113	More vs. Less (ever diagnosed with asthma) 0.79 (0.74, 0.84) I^2 = 71.4% More vs. Less (asthma symptoms within 12mo) 0.76 (0.67, 0.86) I^2 = 91.6%	Age (0-2, 3-6, >=7), design, country affluence, study quality, study year	Kramer Criteria low: 56%, medium: 35%, high: 14%
Brew 2011, Paediatric and Perinatal Epidemiology	MEDLINE and EMBASE	August 2011	Observational (birth cohort and non-cohort) of children 5-18 years age at follow-up with breastfeeding data of 'any', 'none', or exclusive for at least 3-4 months. Included outcomes were current wheezing illness (within the last 12mo).	Current wheezing illness	31	Ever vs. Never 0.97 (0.90, 1.04) I^2 = 63% Exclusive >3-4 vs. <3-4 months 0.96 (0.86, 1.06) I^2 = 54%	Age (5-8, 8-18), Study type, adjustment for confounding, feeding recall, retention rate, country affluence, definition of wheezing illness	Study-specific criteria Ever vs. Never - very low or low: 61% Exclusive - very low or low: 54%

*Definitions of Ever vs. Never and More vs. Less breastfeeding were equivalent between studies and consistent with definitions used in the current review

Table E-2. Search terms and results of database searches

	PubMed	CINAHL	EMBASE	ProQuest Nursing/Allied Health	Summary
Search Details	Search terms: (breastfe* OR breast fe* OR "infant nutrition" OR "breast milk" OR "human milk" OR wean* OR bottle fe*) AND (asthma OR wheez* OR bronchospasm OR "bronchial hyperreactivity" OR "bronchus hyperreactivity" OR "bronchial hypersensitivity") filters: human children 0-18 english result: 985 articles	Search terms: (breastfe* OR breast fe* OR "infant nutrition" OR "breast milk" OR "human milk" OR wean* OR bottle fe*) AND (asthma OR wheez* OR bronchospasm OR "bronchial hyperreactivity" OR "bronchus hyperreactivity" OR "bronchial hypersensitivity") filters: english result: 333	Search terms: (breastfe* OR breast fe* OR "infant nutrition" OR "breast milk" OR "human milk" OR wean* OR bottle fe*) AND (asthma OR wheez* OR bronchospasm OR "bronchial hyperreactivity" OR "bronchus hyperreactivity" OR "bronchial hypersensitivity") filters: human English Child of unspecified age result: 973	Search terms: (breastfe* OR breast fe* OR "infant nutrition" OR "breast milk" OR "human milk" OR wean* OR bottle fe*) AND (asthma OR wheez* OR bronchospasm OR "bronchial hyperreactivity" OR "bronchus hyperreactivity" OR "bronchial hypersensitivity") filters: Adolescent (13-18 years), Child (6-12 years), Infant (1-23 months), Newborn (birth to 1 month), Preschool child (2-5 years) English result: 305	Search Results June 2018: Total: 2596 (from 4 databases) After removal of Duplicates: 2096 Renewal Search (June 2018-Jan 2020) Pubmed, CINAHL, EMBASE and ProQuest Nursing/Allied Health were re-searched for the above timeframe a total of 473 articles (50, 24 and 169 + 230) – 429 after duplicate removal

Table E-3. Summary of study characteristics

Article	Cohort	Design	Country	Study Period	Cohort Size	Age (years)	Breastfeeding Type (cut off)+	Groups included in analyses (months)	Effect Size Measure*	Outcome**	Quality
Ajetunmobi et al. 2015[35]	National Health Service, National Services Scotland	Retrospective	Scotland	1997-2009	502948	0-15	exclusive (1-2 months)	None	HR	Hospitalization discharge diagnosis	Fair
Alm et al. 2008[36]	Infants of Western Sweden	Prospective	Sweden	2003-2004	4921	1	partial (9 months)	>9 vs 5-8 >9 vs 0-4	OR	Criteria: wheezing disorder and inhaled corticosteroids (ICS) use	Fair
Bacopoulou et al. 2009[51]	Greek National Perinatal Survey	Prospective	Greece	1983-2001	6643	0-7	exclusive (3 months)	>3 vs 0	OR	Physician diagnosis and treatment	Low
Bion et al. 2016[21]	Isle of Wight Cohort The Food Allergy and Intolerance Research Cohort	Prospective	England	1989-2008 2001-2011	1087 460	10, 18 10	exclusive, partial (4,6 months)	>6 vs 0 <6 vs 0 4 vs 0 <4 vs 0	RR	Physician diagnosis and wheeze or treatment in past 12 months	Good
Burr et al. 1993[52]	*No name*	Prospective	Wales	1982-1989	452	0-7	ever	None	Pr	Study pediatrician diagnosis	Low
Celind et al. 2018[41]	*No name*	Prospective	Sweden	2003-2015	2433	12	partial (4 months)	>4 vs <4	OR	Physician diagnosis and wheeze or treatment in past 12 months	Fair
Chiu et al. 2016[22]	Prediction of Allergies in Taiwanese Children	Prospective	Taiwan	2007-2010	138	2, 3 , 4	partial (6 months)	>6 vs <6	OR	GINA guideline and either recurrent wheeze or current treatment with confirmation by study pediatric pulmonologist	Good
Davidson et al. 2010[57]	Oxford Record Linkage Study	Retrospective	England	1970-1989	4017	2-11+	ever	None	Pr	Hospitalization with ICD codes	Low
den Dekker et al. 2016[42]	Generation R Study	Prospective	Netherlands	2002-ongoing	4093	0-6	exclusive, partial (4,6 months), ever, per month	None	OR	Physician diagnosis and wheeze in past 12 months	Fair
Elliot et al. 2008[58]	Avon Longitudinal Study of Parents and Children	Prospective	England	1991-ongoing	8131	7.5	exclusive (4 months)	>4 vs <4	OR	Physician diagnosis and either wheeze or treatment in past 12 months	Low
Fredriksson et al. 2007[43]	*No name*	Prospective	Finland	1984-1997	385	7-14	partial (multiple 3-12 months)	4-6 vs 0-3 7-9 vs 4-6 10-12 vs 4-6 >12 vs 4-6	OR, Pr	Physician diagnosis	Fair
Karmaus et al. 2008[31]	Isle of Wight Cohort	Prospective	England	1989-2008	1342	1-10	partial (3 months)	>3 vs <3	RR	Physician diagnosis and wheeze in past 12 months	Good

Kiechl-Kohlendorfer et al. 2007[44]	Survey of Tyrol, Austria	Prospective	Austria	1994-2005	33808	5-11	exclusive, partial (1-2 months)	None	OR, Pr	Hospitalization with ICD codes and 2 study pediatric pulmonologist confirmation	Fair
Klingberg et al. 2019[32]	All Babies In Southeast Sweden Cohort	Prospective	Sweden	1997-2014	9286	0-17	partial (9-10 months)	>9.75 vs <7.5	OR	ICD codes for atopic/non-atopic asthma	Good
Klopp et al. 2017[33]	Canadian Healthy Infant Longitudinal Development Study	Prospective	Canada	2009-2015	2534	3	exclusive (3 months), ever	>3 vs 0	OR, Pr	Possible/probably asthma diagnosis by medical professional	Good
Kull et al. 2002[45]	Children, Allergy, Milieu, Stockholm Epidemiological Cohort	Prospective	Sweden	1994-1996	321	0-2	exclusive (4 months), partial (6 months)	>6 vs <6 >4 vs <4	OR, Pr	Criteria: either 3 wheeze episodes and ICS, or hyperreactivity without signs of respiratory infection	Fair
Lee et al. 2017[34]	*No name*	Prospective	Taiwan	1998-2009	737	3-6	partial (1 month)	>1 vs <1	OR	Physician diagnosis with exercise induced cough and/or night cough >2 weeks in past 6 months	Good
Leung et al. 2016[23]	Children of 1997, Hong Kong Chinese Birth Cohort	Prospective	Hong Kong	1997-2012	8301	0.25-12	exclusive, partial (3 months)	None	HR	Hospitalization based on ICD codes	Good
Mandhane et al. 2007[59]	Dunedin multidisciplinary health and development research study	Prospective	New Zealand	1972-1998	418	9	partial (1 month), ever	>1 vs 0	OR, Pr	Current wheeze and methacholine challenge	Low
Mcconnochie et al. 1986[60]	*No name*	Retrospective	USA	1971-1981	224	0-10	ever	>0 vs <0	RR, Pr	Physician diagnosis and treatment	Low
Midodzi et al. 2010[24]	Canadian Early Childhood Development cohort	Prospective	Canada	1996-2003	8499	2-5	partial (3 months)	>3 vs 0 <3 vs 0	HR, Pr	Physician diagnosis	Good
Midwinter et al. 1987[61]	*No name*	Prospective	England	1979-1984	367	0-5	exclusive (1 month), Ever	None	Pr	Physician diagnosis	Low
Mihrshahi et al. 2007[46]	*No name*	Prospective	Australia	1997-1999	516	5	partial (3,6 months), ever	>6 vs <6 >3 vs <3 >0 vs <0	OR, Pr	Wheeze in past 12 months with previous diagnosis, hospitalization or positive bronchodilator test	Fair
Milner et al. 2004[62]	National Maternal and Infant Health Survey	Retrospective	USA	1988-1991	8071	0-3	ever	>0 vs <0	OR, Pr	Diagnosis by medical professional	Low
Miyake et al. 2008[47]	Osaka Maternal and Child Health Study	Prospective	Japan	2001-2005	763	0-2	Exclusive (4 months) Partial (6 months)	>4 vs <4 >6 vs <6	OR, Pr	Physician diagnosis	Fair
Nwaru et al. 2013 (1)[25]	Study of Eczema and Asthma to Observe the influence of Nutrition	Prospective	Scotland	1997-2008	879	0-10	exclusive, partial (2-3, 3-4 months), ever	>3.75 vs 0 <3.75 vs 0	OR	Physician diagnosis	Good
Nwaru et al.	Finnish Type 1 Diabetes	Prospective	Finland	1994-1999	2617	5	partial	9.5 vs <5	HR	Physician diagnosis and	Low

2013 (2)[53]	Prediction and Prevention						(9-10 months)	9.5 vs 5.0-9.5		either wheeze or asthma treatment in past 12 months	
Oddy et al. 2002[27]	Western Australian Pregnancy Cohort	Prospective	Australia	1989-1996	1980	6	exclusive (4 months)	>4 vs <4	OR, Pr	Physician diagnosis and wheeze in past 12 months	Good
Oddy et al. 2004[28]	Western Australian Pregnancy Cohort	Prospective	Australia	1989-1996	1485	6	per month	None	OR	Physician diagnosis and wheeze in past 12 months	Good
Sbihi et al. 2016[54]	The Border Air Quality Study	Retrospective	Canada	1999-2009	65448	0-10	ever	>0 vs 0	OR	Physician billing/discharge ICD codes	Low
Silvers et al. 2009[29]	New Zealand Asthma and Allergy Cohort Study	Prospective	New Zealand	1997-2002	1011	0-1.25	per month	None	OR	Physician diagnosis and wheeze in past 12 months and inhaler	Good
Silvers et al. 2012[26]	New Zealand Asthma and Allergy Cohort Study	Prospective	New Zealand	1997-2007	999	2	per month	None	OR	Physician diagnosis and wheeze in 12 months and inhaler	Good
Standl et al. 2012[55]	GINIplus & LISAplus	Prospective	Germany	1995-2008 1997-2009	1691	0.5-10	exclusive (multiple 1-6 months)	None	Pr	Physician diagnosis with IgE for atopic asthma	Low
Sunyer et al. 2006[48]	Menorca, Spain Cohort Study	Prospective	Spain	1997-2004	332	0-6.5	ever	>0 vs 0	OR	Physician diagnosis	Fair
Turner et al. 2008[49]	*No name*	Prospective	Australia	N/A	253	2-11	exclusive (6 months)	>6 vs <6	OR	Physician diagnosis	Fair
van Meel et al. 2017[30]	Generation R Study	Prospective	Sweden	2002-2016	4464	10	exclusive, partial (6 months), ever, per month	>6 vs 0 >6 vs <2-4 >6 vs 4-6	OR	Physician diagnosis and either wheeze or treatment in past 12 months	Good
Von Kobyletzki et al. 2012[50]	Dampness in Building and Health (DBH)	Prospective	Sweden	2000-2005	3017	1-5	partial	>6 vs <6	Pr	Physician diagnosis	Fair
Wickman et al. 2003[37]	Children, Allergy, Milieu, Stockholm Epidemiological Cohort	Prospective	Sweden	1994-1998	3692	0.25-2	exclusive	>6 vs <3 4-5 vs <3	OR, Pr	Criteria: 3 wheeze episodes and either ICS or hyperreactivity without signs of respiratory infection	Fair
Wilson et al. 1998[38]	Dundee Infant Feeding Study	Prospective	Scotland	1983-1993	545	0-10	exclusive, partial (3-4 months), ever	None	Pr	Physician diagnosis or current asthma treatment	Fair
Wright et al. 2000[39]	Tucson Children's Respiratory Study	Prospective	USA	1980-1995	792	0-11	exclusive (4 months)	>4 vs <4	OR	Physician diagnosis and wheeze in the past 12 months	Fair
Wright et al. 2001[56]	Tucson Children's Respiratory Study	Prospective	USA	1980-ongoing	926	6-13	exclusive (4 months), ever	None	Pr	Physician dx + either wheeze or asthma symptoms reported on 2+	Low

										questionnaires	
Yamakawa et al. 2015[40]	Longitudinal Survey of Babies in the 21st Century	Prospective	Japan	2001-2008	13853	0.5-3.5	exclusive, partial (multiple 1-7 months), ever, per month	None	OR	Hospitalization in past 12 months for asthma	Fair

*HR = hazard ratio, OR = odds ratio, Pr = Prevalence

** Outcome definitions have been paraphrased for simplicity